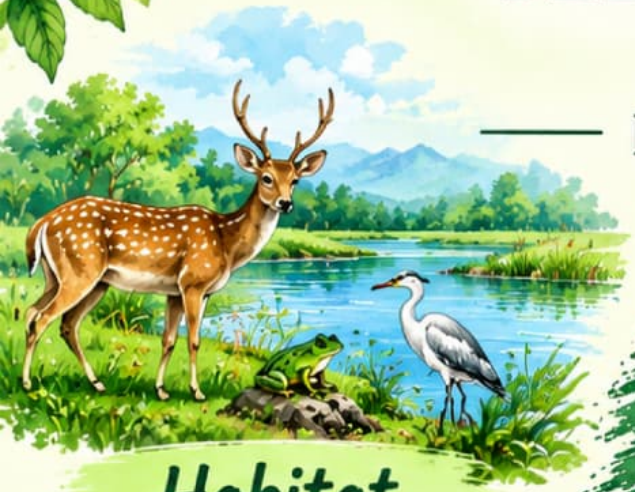
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Habitat and Niche



Habitat
The natural place where an organism lives.



Niche
The role and function of an organism in its ecosystem.



B.Sc. 2nd Semester | CBPBU

Created by Bikram Singha

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Habitat & Niche

9) Describe different types of niche concept in Ecology.

An ecological niche refers to the role and position a species has in its environment. It includes

- How it obtains its food
- its behaviour & adaptation
- How it interacts with other organism
- How it contributes to energy flow & nutrient cycle in an ecosystem

The niche concept

- first proposed by → Russian ecologist → Georgy Gause → in 1930
- introduced the idea of the niche as – The ecological role of a species within its community.

- A niche is based on → position of an organism in the ecosystem
- There are four aspects of an ecological niche–
  - Special or Habitat niche
  - Functional or Tropical niche
  - Multifunctional/ n-dimensional/ Hypervolume niche
  - modern ecological niche

- Core components of an ecological niche

An ecological niche is determined by both biotic (living) and abiotic (non living) factor. Includes

Resource utilization	Functional Role	Timing	Abiotic require-ments
> What the organism eats (food),	> Its position in the food chain eg- producer, consumer, Decomposer	> when the organism is active eg- nocturnal Diurnal	> Temp^{r} tolerance
> Where to get water			> moisture level
> Material it uses for shelter			> Soil
			> nutrients required.

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● Four aspects of ecological niche –

1) Spatial niche or Habitat niche –

→ A spatial niche refers to the specific physical space, location, or microhabitat occupied by an organism within an ecosystem. (to survive & avoid competition)

→ Spatial niche → also known as the habitat niche.
↓
proposed by → Grinnell
↓
According to Him –
'Niche is the area where an organism live'.

→ Examples of spatial niche

① Forest birds
> Some birds ~~live~~ live at the top of trees (canopy); while others stay in lower branches or near the ground.
> Same forest, they use different spaces.

② Fish in a lake
> Certain fish stay near the surface, where other live in deep water
> Different depths = different spatial niche.

③ Plants in a forest
> Tall trees grow in sunlight while small plants grow in the shade of trees.
> Different vertical spaces.

④ Lizards on a tree
> one sp. stays on tree trunks, another on leaves, another on branches
> Same tree different spaces.

2) Functional or Tropical niche –

→ Tropical niche is concerned with the tropical position of an organism.

→ Tropical niche → also known as the Ellonian niche
↓
proposed by → Odum

→ Examples of Tropical niche

① Two aquatic bugs Notonecta (an active aquatic predator) and Corixa (detrivor) live in same pond in the littoral zone of the pond but their food source are different.

② various species of finches of galapagos island belong to the three genera viz Geospiza, Camarhychus and certhidia – all live in same general habitat but differ in trophic positions.

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Examples of Trophkal niche

- Food Source differentiation
  - Notonecta (backswimmers) → act as predator → feeding on other animals
  - Corixa (water boatman) → act as detrivores → feeding on decomposing organic matter
  - Both live in same pond → Littoral zone of the pond.
- Specialized feeding
  - Galapagos finches / Darwin's finches
  - 14-15 species evolved from a common ancestor
    - Geospiza → Seed eaters
    - Camarhynchus → fruits & buds eaters
    - Certhidea → insect eaters (insectivores)
  - Different tropic position but live in same habitat → To avoid competition by the help of resource partition

3) Multifunctional / n-dimensional or Hypervolume niche

→ This is a concept of ecology ~~that~~ that describes the set of environmental conditions and resources within which a species can survive and reproduce.

→ multifunctional niche → proposed by → G.E. Hutchinson

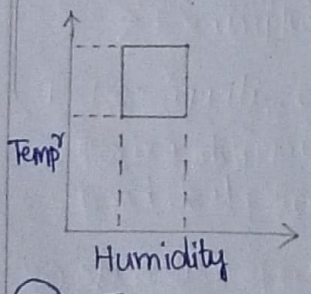
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According to him

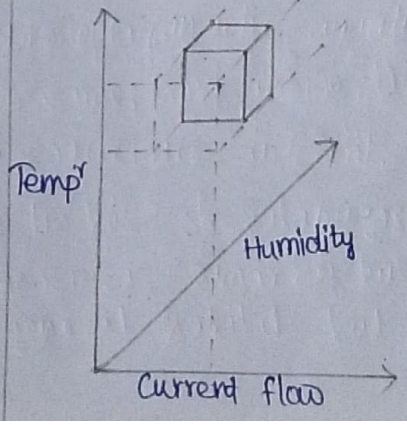
↓

∅ The organisms perform its functional role under environmental or certain condition.

∅ & Every organisms depends on environmental gradient for survival → such as temp^r, light, moisture, height of vegetation etc.



(2D) → 2D graph is never possible in nature



→ 3-D → According to Hutchinson

↓

A large number of biotic and abiotic factors affects the functioning of a population. Hence the niche is hypervolu or - n-dimensional

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4) Modern ecological niche -

proposed by
↓
Soberon & Peterson

Considers the access of an organism or parasite to both biotic and abiotic factors that allow it to survive.

— This include environmental conditions & how an organism interacts with its surroundings.

ø Niche width and Niche overlap.

Niche width

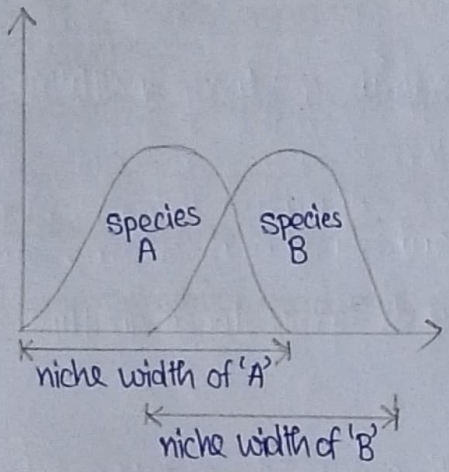
Refers to the range of resources or environmental conditions a species utilizes,

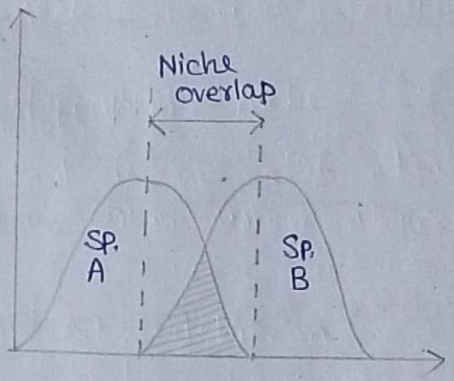
or

the range of conditions tolerated by the individuals in the population.

Niche Overlap.

Refers to the extent of similarity in resource utilization or tolerance of conditions.





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Ø Ecotone :-

1) What is ecotone?
2) Why is ecotone formed?
3) Features of ecotone?
4) Types of ecotone.
5) Examples of ecotone.
6) ~~Ex~~ importance of ecotone.
7) Ecotone v/s Edge v/s Boundary.
8) Why species diversity is high in ecotone?
9) Ecological importance of ecotone.

1) Ecotone-

→ It is a transitional zone b/w two ~~ecosystem~~ different ecosystem or communities.
→ It is a zone of junction where two communities meet and integrate
→ It consists organisms from both ecosystems and some unique species.
→ example- forest - grassland boundary.

2) Ecotone forms due to- (BEST Human)

a) Environmental gradients - like change in temp^r, light, moisture

one ecosystem slowly changes into another

example → forest ⟶ Grassland
(cool, shaded) (open, sunny)

b) Soil differences -
→ change in soil type, pH, nutrients, or moisture creates different plant ~~comm~~ communities.
→ where two soil type meet, an ecotone forms,

c) Topography - (Land shape)
→ change in height, ~~so~~ slope or altitude affect climate & vegetation
ex- Hill slope ⟶ Plain land transition.

d) Biological Interaction -
→ Interaction b/w species like- Competition, migration
→ Species from both ecosystem mix & create a transition zone

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e) Human Activities –

→ Activities like ↓

Deforestation
Agriculture
Urbanization

→ create artificial ecotones
eg- framland next to forest.

3) Features / characteristics of Ecotone –

Ecotone has some special characteristics that make it different from normal ecosystems.

1) High Species diversity –

Ecotone shows very high biodiversity. Because it contains

> Species from both ecosystems
> Some unique species

This makes it richer than either ecosystem alone.

2) Presence of Ecotonal (Unique) species –

> Some species are found only in ecotone, not in adjacent ecosystems.

> These are called ecotonal species → They are specially adapted to changing conditions.

3) Edge effect –

> Ecotone shows increased population density and species richness.

> Reason –
- more food availability
- more shelter
- mixed habitat conditions.

4) High Competition –

> Species from two ecosystems live together. So, there is intense competition for food, space, light → leads to natural selection & adaptation.

5) Grater Productivity –

Ecotone often have higher biological productivity.

Due to ↓

Better resource availability

Favorable conditions for growth

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6) Sensitive to environmental Changes

> Ecotone is very sensitive to

Climate change | pollution | Human disturbance

> Small changes can affect it quickly.

7) Acts as a Buffer zone

> It acts as a buffer or transition area b/w two ecosystem.
> Helps to reduce sudden environmental changes.

8) Dynamic Nature

> Ecotone is not stable, it changes over time
> Boundaries may shift due to
  > Climate change
  > Human activity
  > Natural succession.

4) Types of Ecotone

Natural ecotone
↓
formed naturally
ex→ forest – grassland

Artificial ecotone
↓
Created by humans
ex→ Agricultural land near forest.

5) Examples of Ecotone

1) Forest – grassland boundary
2) Riverbank (Land–water)
3) Estuary (Freshwater – saltwater)
4) Mangrove (marine – terrestial ecosystem)
5) Grassland (b/w forest – desert)

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6) Importance of Ecotone

1) Supports high biodiversity.
2) provides habitat for many species.
3) Helps in ecological balance
4) Acts as an indicator of environmental change
5) Important for wildlife conservation.

7) Ecotone v/s Edge v/s Boundary

Feature	Ecotone	Edge	Boundary
Meaning	Transition zone	margin of ecotone	Sharp separation
width	wide	narrow	very narrow
Species diversity	High	moderate	Low
Example	Forest–grassland	Border line area	wall-like separation

8) Species diversity is high in ecotone as-

1) Species from both ecosystems are ⊕
2) more food & resources available.
3) More habitat variety.
4) ⊕ of ecotonal species.
5) Due to edge effect.

9) Ecological importance of ecotone-

1) Increase biodiversity.
2) provides shelter & breeding ground for organisms.
3) Helps in energy flow and nutrient cycling.
4) Acts as buffer zone b/w ecosystems.
5) Useful in environmental monitoring.

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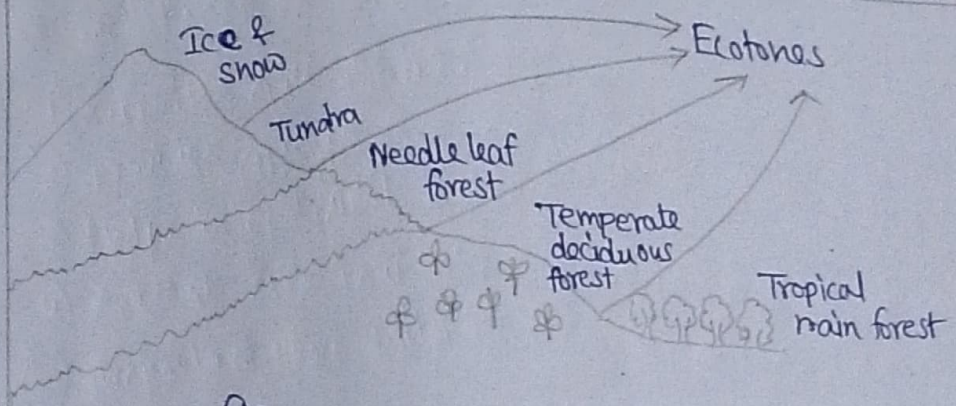
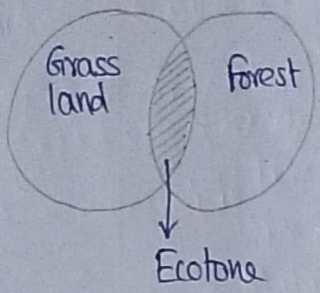


Fig -



Edge effect

> A general characteristic feature of the ecotone is that it has sufficiently greater number of species and moreover the density of most of the species is higher than that in the neighbouring communities, it is called edge effect.

> The organisms which occur primarly or most abundantly in the ecotone are known as edge species.

Explanation

1) In an ecotone, organisms from both ecosystems are present.
2) Also some unique species (ecotonal species) are found only in this area.
3) Due to mixed environmental conditions, the area becomes more suitable for many organisms.

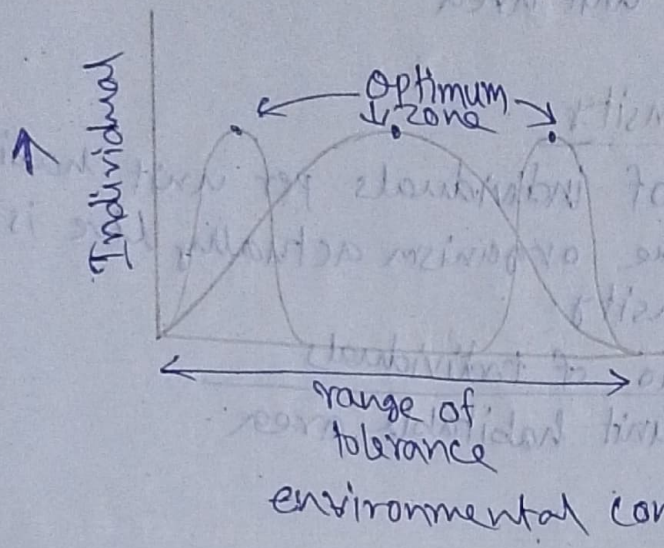
Cause of edge effect

1) Availability of more food and resources.
2) presence of different habitats together.
3) Favourable environmental conditions.

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Ø Ecological Amplitude-

Every species differ from the range of environmental conditions such as pH, salinity, temperature. ~~The~~ Thus the range of tolerance of a species is called as ecological Amplitude



They are of Two types

Eury
↓
long range of Tolerance

ex → Eury-therm
↓
long range of Tempr tolerance

Steno
↓
short range of tolerance

ex → Steno-therm
↓
short range of Tempr tolerance

If a ~~new~~ species bring to a new habitat-

① Ecophene-
- The species will change its morphological appearance to survive under that condition
- ~~Temporry~~, Temporary, non-genetic

② Ecotype-
small minute genetic change

③ Ecospecies-
Large genetic change & become new species

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Crude density-

The number of individuals per unit area (habitable & non habitable both) is c/o crude density.

$$CD = \frac{\text{no of individuals}}{\text{unit area}}$$

Ecological density

The no. of individuals per unit habitable area where the organism actually live is c/o ecological density.

$$ED = \frac{\text{no of individuals}}{\text{unit habitable area}}$$

Epillimnion-

Epillimnion is the upper warm layer of water in a stratified lake that recieves sulight well

eg- phytoplankton, zooplankton, Diatom found in that area

Hypolimnion

Hypolimnion is the lower cold water layer where sulight ~~no~~ donot reach.

ex- Bacteria, fungi mainly decomposer found in that area.